Getting the Reactor to Work - Possible Final Edit by Wesley Savage:

This is a collection of key items to getting the
reactor to reform fuels and not just vaporize them. The GEET "Free Internet Plans" have been described by Paul Pantone as the worst design in 25 years so please consider one of the plans below, the Suppemental Addendum to the Small Plans, and read this file's text to see how to make the GEET work better.

10_to_25hp_GEET_Reactor_Plans.gif

For smaller engines:
GRVOFSR1.zip
GRVOFSR2.zip streamlined flow, bigger sight glasses
GRVOFSR3.zip economy version, about $40US.
UPDATE:
This zip's PDF shows an easier way to make a GEET. This is for about a 1hp engine but this same way of
making the GEET can be scaled up if you want to make a larger GEET:
GRVOFSR5.zip smaller version, easier to construct.

1. Reactor Magnetics:

Use ferromagnetic pipes and rod, like steel or iron. Some ferromagnetic materials or coatings work even better but still must have some iron. Mike Holler, previously GEET R&D person and president of GEET, would not say which ones work better but said some do. Stainless steel has worked for some people after it was heated from use and developed a remnant magnetic field. In fact it appeared to work even better than regular steel. Also, some grades of stainless steel (possibly the poorer quality grades) work better for retaining remnant magnetic fields after initial use. Mild steel can develop a stronger magnetic field during use than stainless steel. I'm not sure if stronger is better but definitely the pipes should retain remnant magnetic fields of the correct orientation based on opposites attract. This leaves us with Mild Steels, Black Iron, Duplex Stainless Steel, and Ferritic Stainless Steel to work with.

If possible, the tees and elbows and unions should be demagnetized, which can be done by heating them up above their Curie temperature i.e. usually until just barely starting to turn red in a dark room and then quench the parts in water. After the reactor has been run with an engine you can use a compass to check that the magnetic orientations of pipes and rod have developed like in the file GEET Reactor Magnetics gif or the other images files I've made myself.

2. Rod Length and Shape:
Make your rod length to match the reactor orientation and fuel type. If the reactor is oriented vertically as opposed to horizontally it will affect the length of rod that makes the reactor work the best. Paul Pantone recommends these rod lengths based on fuel type and reactor orientation:

water as fuel - 32mm vertical rod or 64mm horizontal rod
petrol - 92mm vertical rod or 184mm horizontal rod
diesel fuel - 115mm vertical rod or 229mm horizontal rod
crude oil - 156mm vertical rod or 305mm horizontal rod

If the rod is a too long (for the fuel in use + the reactor's orientation) it will not be as efficient as it could be but if the rod is too short (for the fuel in use + the reactor's orientation) then the reaction will not start at all and it will only act like a fuel vaporizer instead of a fuel reformer. Unfortunately, gap size (see 8 below) and the exact type of ferrous metal used for the pipes and rod affect the correct length to use also.

If you start with a rod too long and burn it in to get a magnetic signature on it, then Mike Hollar (previously president of GEET) says do not cut that rod to the length determined by where compass shows the ends to be but rather cut a new rod to length based on the end points of the burned in rod. But he never said why to use a new rod, but I think there's at least a reasons why.

Firstly, you can use the old rod to make new rods the same length as the burn-in patterns on it and in proper north and south points revealed by the compass and shapes of each end being properly done.

Secondly, it's probably best to burn a new rod already cut and shaped of that exact length in to tune the reformer to ensure it's gonna run as you desire after burning it in for any length of time from 30 minutes to a couple hours.

Paul Pantone says: "...after a "burn-in" period of at least thirty to sixty minutes, remove the rod, using a small compass, move the compass over the rod from one end to the other. The compass will swing from one direction to the other at the two points that the rod needs to be cut at for optimum development of the static-electric field that is required to develop to induce/start the plasma discharge field at both ends of the rod. Remember, the tolerance between the rod and the reactor housing must be kept to a minimum for the air-flow (CFM) that the engine requires for the operation range that it is to be used for, so that the rod hovers in the airflow, thus not shorting out to the reactor housing walls and stopping the plasma field discharge. Remember to put a .250 inch diameter point on the south end of the rod (reactor inlet end) with a 17 deg taper to the rods full size and hollow grind the north end (concave grind)."

I think a .25" diameter point means a .25 inch rounded ball or nose to the rod. Some people have had to run their engine+GEET for more like 10 hours before starting to get a good reaction. When Paul talks about the rod hovering so as not to short out, it is best to use a rod with no centering posts on it.

3. Pipe Lengths:
Make the area where inner and outer pipes overlap no more than an inch longer max at each end than the rod length. Preheating of the vapors, as occurs in longer pipes before and after the rod's position, is taking away from the needed temperature differentials as the vapors swirl past the rod. The pipes don't have to be like in the diagram linked to in 1. Instead of just straight pipes, the pipes could also be L shaped or U shaped but the area of pipe inside pipe where heat exchange occurs should not be very much longer than the rod length.

4. Reactor Vacuum:
UPDATE: Having a good vacuum is likely more important for creating a fast flow of gases around the rod which in turn creates a substantial venturi vacuum increase just over the area where the vapours speed up past the rod. If the reactor's input gases have a high flow rate and if the fuel and water are sufficiently vaporized before they go into the reactor then the reactor may still work well with less applied vacuum since the vapours still get an additional venturi vacuum boost as they go past the rod.

OLD TEXT of section 4:
Insure that where your vapors go into the reactor to get reformed you have at least 8 to 10 inches of vacuum. Like is listed in the 4th row of this chart where it shows 10.24 inches HgV mercury vacuum:

http://www.pauloabbe.com/productLines/
vacuumDryersSystems/vacuumconversionchart.html

I'm talking about between the bubbler and the reactor's input, or between the fuel injectors or carburetors used for the input of water and fuel vapors to the reactor. At maximum flow rate for high engine RPM the input end of the reactor needs 8 to 10 inches of vacuum, 5 at the very least. Gases can stay in a plasma state more easily in a partial vacuum. That is how a fluorescent light works. In the reactor the gases are in a partial vacuum and then electrostatic charge buildup in the gases keeps them ionized. This means that the fresh air opening, for mixing reactor gases with fresh air, needs to be smaller than the original carburetor air intake opening so the greater restriction will keep the vacuum as much as 8 to 10 inches at full throttle based on Bernoulli's Principle and the Venturi Effect. The amount of reactor gases must be controlled with a needle valve or similar metering valve or a small vacuum relief valve inline with the gases going into the reactor. Where fresh air comes in to mix with reactor output gases you need a vacuum relief valve that shuts off more fresh air when the vacuum to the reactor is not enough. A spring loaded choke plate or a commercial adjustable relief valve can be placed up stream.

5. Creating the Cool Water Vapours plus Fuel vapors:

A bubbler container with a fog nozzle works better to make a dry fog of droplets about 10 micron in size. oily fuels never vaporize. You can send in a fog or vaporize water and the water in a bubbler tends to carry oily fuel molecules in with it. Usually there is so much water vapor in the exhaust that a bubbler for water is not required, just a very small carburetor or a fogger nozzle with a high pressure pump and then send through exhaust gases to draw out the atomized fuel. However, it can still be helpful to have the fresh air go into a bubbler of water with electrolye and then into the reactor along with the vapors from the fuel source. Fuel and water must get turned into an aerosol from bubbling agitation, not by heating until evaporation. Please ask a lot of questions in the group about getting dry vapors into the reactor because this is what many have trouble with.

Use a bubbler container that will not collapse in under a strong vacuum. The vacuum reservoir cans found in cars use inward indentions around the circumference to give the can more strength. Whatever you use for a bubbler, insure it will not collapse under high vacuum.

Insure that there are many small bubbles coming from the bubbling device in the bubbler container. I have used 1/8th NPT pipe down to a metal speaker grill with porous landscaping cloth under it.

A stainless steel pipe and grill would be better for preventing corrosion. Then stainless steel wool can be used to help break up the large droplets. The bubbling needs to keep the heavier fuels churned up so they don't just settle to the bottom.

Others have tried injectors that squirt fuel directly on a hot pipe heated with exhaust gases. This can work if you insure you get as much heat out of the exhaust as you can with a very good heat exchange setup. An aerosol of cool water mist should be added to hot vaporized oil because the mix must be cool as it enters the GEET.

Very small carburetors can be used also. If the GEET reactor is too large for the size of engine then more fresh air entering the reactor along with the fuel and water vapors will dilute the vapors to be reformed and help keep a fast flow rate through the GEET. Paul Pantone used to say to make very rich vapors to send into the reactor, however later when Mike Holler was doing R&D work for Paul, he found out that he got better performance from the reactor when he diluted the vapours with more fresh air because that allowed him to get a faster flow rate.

6. Cool and Dry (or not?) vapors:

In a GEET training video, Mike Holler says to insure the vapors going into the reactor are as dry as you can get them. I believe his primary motivation for saying this was the good results of one experimenter can get them. I believe his primary motivation for saying this was the good results of one experimenter who was also playing with catalyst coatings and he had good results with a very dry vapor.

On the other hand, if you look at the Paul Pantone's video demo of "Old Blue" you see he is not doing anything special to create super dry vapors. One thing is for sure, too wet of a mist can kill the reaction. Having said that, I should also say that creating the most electrostatic buildup around the rod is very important. Within thunderstorms, it is while water is changing phases between a gas and a liquid that the greatest electrostatic build up appears to occur. I believe the same is true when going from a liquid to a gas state. If true, then a very fine fog of fuel + water transitioning to a dry vapor while swirling past the rod should create the most electrostatic buildup. In any case the rule still applies that too wet of a mist will kill the reaction.

Also, get the vapors into the reactor as cool as you can. If you have 8 to 10 inches of vacuum it should help stop the vapors from condensing. But that applied vacuum is not as important as the venturi vacuum that develops in the gap past the rod from the vapors speeding up as they go past the rod. The cooler the vapors are as they enter the GEET the better the reaction.

7. Rod Freedom to Center Itself:

Insure the rod is free to move around in the end to end direction. You need some loops of wire at the ends to stop the rod from going too far but still the rod needs a good range to move on its own. A horizontal or tilted orientation of the reactor may work better for this than a vertical orientation. I need to hear feedback from more experimenters about if they have just as much success running with shorter rods in a vertical orientation.

Paul Pantone made several demo units in which the reactor chamber is mostly horizontal but the end with the rod's pointing end is oriented downward to some degree. The gases going into the inner pipe tend to blow the rod up the pipe but gravity tends to make the rod fall down the pipe and the 2 forces are somewhat balanced at a somewhat downward orientation and this helps the rod to hover at the best location in the reactor's inner pipe.

Once the circulating vapors are creating their own magnetic field, that magnetic field will pull the rod forward into the incoming vortexing gases. This is similar an electric solenoid coil that pulls its center rod into it when an electric current energizes the coil of wire. But in this case, the circulating ionizedn gases act as a coil of electrically conducting wire in a solenoid. The magnetic properties of both the rod and the pipes determines where the rod gets pulled to once a magnetic field is created by circulating ionized gases.

Rods with no cenering posts and are as smooth as possible will work better overall and will work better ESPECIALLY if the engine RPMs are in the mid to high range because they can float and spin in the vortex better than rods with centering posts.

8. Rod in Pipe Gap Size:

Insure you have a small gap size between the rod and the inner pipe's inner wall. For engines smaller than a car engine a 1/32 inch or 1mm gap or even a smaller gap might be good but on small GEET reactors I've seen them work well with up to an 1/8th inch gap all around the rod si I'm not sure what the best gap size is. Paul likes to keep that proprietary. For a reactor to go with a car engine a 1/16 inch or 1.5 to 2mm gap may work. Paul Pantone knows the best gap sizes and this is apparently an important part of getting a really strong reaction since Paul wants to give out the information only in his classes or to people who have bought his plans.

If you have larger gap sizes then send much more fresh air gases through the reactor along with the fuel and water vapors so that the flow rate of the vortexing vapours stays high.

Paul Pantone once said he found that he did not have to keep making the reactor bigger and bigger for using with very large engines.

Finally, you use must steel pipes or tubes with a very smooth inner wall, with no weld ridge. I've heard and SEEN that properly made black iron (magnetite plated) pipe nipple made in the USA and sold by MSC Industrial Supply Company has a smooth inner weld seam but they still have a rough surface on the inside. The pipes or tubes must be very smooth inside. User a reamer to make pipes very smooth inside else use DOM steel, or ferritic or duplex stainless steels for the tubing instead of pipe.

9. Using Electrolytes:
Before explaining why using electrolytes can be important, please review what electrolytes are:

http://en.wikipedia.org/wiki/Electrolyte

At that webpage, pay attention to the sentence that says:

"Furthermore, some gases may act as electrolytes under conditions of high temperature or low pressure."

When electrolytes are vaporized in the bubbler or with a carburetor then there are more ions in the vapor. As these ions move through the magnetic field between the inner pipe and the rod, the ions get electromagnetically deflected to a circular path. The +ion will get deflected 1 direction and the -ions the other direction. One polarity of ions will get magnetically deflected to swirl in a vortex closer to the rod while the other polarity of ions will tend to swirl in a vortex closer to the inner pipe wall, and they will swap sides as they go through the area where the magnetic field direction changes. During this transition there can be collisions between them and this may be part of the phenomena occuring but not yet fully, but are akin to particle accelerator collisions from what's little understood.

With both sets of ions, their flow constitutes an electric current flow that creates a stronger magnetic field and these ions moving through a stronger magnetic field makes them swirl more and to get magnetically compressed and as they get magnetically compressed, they get accelerated around the rod even further. This creates an area of increased temperature and that assists in the fuel reforming.

Ammonia molecules are loaded with hydrogen that gets released easily in the reforming process. It is not a coincidence that Paul Pantone had good results using urine mixed with the hydrocarbon fuels. Paul has also used salt water with good success. Some people have cautioned against using salt for the electrolyte because of how corrosive it can be. But Paul had good success using it in his "Old Blue" demo unit. He used an aluminum bubbler container that was less susceptible to salt corrosion than a steel container. However, with some salts, acids and bases, aluminum might be corroded more easily than steel so you need to be careful what you use for an electrolyte. Paul also had success with some sulfuric acid in the salt water and crude oil mix but again, be careful what material your container is made of to be sure it is not corroded by the electrolytes.

UPDATE: Several people have had their GEETs get very corroded inside when using salt in the water when using regular steel pipes. Stainless steel pipes or tubes are much less susceptible to these corrosion problems. Still the engine itself can have corrosion problems if there is nothing to stop corrosive moist vapours from reaching the engine when it is not in use.

10. Surfactants and Emulsifiers:

When a small amount of these are added to a bubbler then it helps oily fuels and water to cling together so both get carried into the GEET by agitation of the bubbles. It is agitation, not heat that makes the aerosol to go into the GEET. Paul Pantone liked Dawn dish washing liquid but I don't know if they have changed their formula over the years. I saw 1 youtube video where that created a clear, non milky, emulsion using these additives:

triton-X100 n-butanol (a fermintation type of alcohol, not rubbing alcohol) and water solution of $La(NO3)3$ (Lanthanum Nitrate, strong oxidizer) That last one looks dangerous and I don't know where to get it. The first one is a component in pet cage cleaner:

http://www.nlspets.com/Clean-Cage-8-Oz-1-p/276355.htm

I have also seen patents for making a microemulsion of fuel and water that use water + fuel + a fatty acid + ammonia or urea or DMAE or nonylphenol ethoxylate. For the fatty acid, olive oil is about 50% oleic acid and safflower oil is almost 80% linoleic acid. Only a small amount of the fatty acid and the emulsifier should be needed but I haven't tried this myself. These are only needed if trying to use a bubbler to consume things like used oil. For petrol and even diesel fuel, people have successfully used a small carburetor to feed the GEET.

11. Making the aerosol droplets as small as possible:

A carburetor tends to make larger droplets than an injector but the droplets can still be made substaintially smaller by sending the carburetor output through a long pointed cone shaped piece of 35 to 40 mesh screen with the point facing away from the carburetor. Don't make the mesh holes too small or it helps the droplets to form back into bigger droplets again. Holes as big as in window screens can work if the angle of incidense with the droplets is good as with a longer cone.

Check out the Cone Screen Image that's in the mega.nz public download as the primary example

12. Smooth Flow Through the Pipes:

The standard threaded pipes used in the plans are actually not the best choice. If you know how to weld it is better to make the whole thing with much smoother transitions and with pipes that are very smooth on the inside. The pipe size should reduce in the area where the exhaust gases pass where the rod is in the inner pipe and then the outer pipe should increase in diameter again. See the 3rd diagram on sheet 5 of the 5 - geetro rev pdf file for a good example.

The flow through the outer pipe needs to be similar flow through a venturi except this one has a pipe through the middle of it. Also, the gases will tend to swirl like water going down the drain and the smoother the pipes the easier it is for the swirl to develop. The swirl motion is very important in both the inner and outer pipes, however the best performance comes from letting the swirl develop in the direction and pitch that it wants rather than trying to force it to a particular direction or pitch.

13. Good Electrical Connection Between Pipes:

Paul Pantone says DO NOT use Teflon tape between pipes and pipe fittings or you will not get a good reaction. I assume this is because there needs to be a very good electrical connection to get a strong reaction. The GEET outer pipe may act as the electrical return path for the current flow in the inner pipe that maintains the plasma. People who have made welded GEETs seem to get stronger reactions. Slightly thicker tubing rather than thin wall tubing may help also to get a good current flow and good reaction inside. The higher the current flow the more intense the plasma inside. One experimenter with a well welded stainless steel GEET has measured over 100 amps of current through the middle of the GEET using a clamp around current probe. Paul also says to keep plastic tubing away from the GEET so it won't meld but the plastic might also affect the energy fields of a well running GEET so as to not get a good reaction.

14. Ensuring Electrostatic Charge and where not to use Teflon tape:

I will say using teflon tape would be just fine for the vapor making stage and the lines going to the tube feeding into the brass nipple fitting that's ideally 7 to 8 inches away from the intake of the inner pipe of the reactor, because teflon tape is a dielectric and if you're using a copper tube like many free online plans use for the intake to the bubbler, copper is conductive and will ground out the electrostatic buildup, so this will minimize or prevent that, because you want the static electricity to build up in the vapors so they will convert to current by electrostatic discharging inside the reactor piping, just remember to KEEP TEFLON TAPE AWAY FROM THE REACTOR PIPING EVERYONE!!

When pursuing matching up the remnant magnetics for building the GEET Reactor and following the Triboelectric Series or Table, remember these rules of thumb to both.

Opposites Attract in Magnetics and Electromagnetics, so opposite magnetic bodies are drawn together.

Opposite Electrical Charges rub to make static electricity.

According to Michael Faraday, J.J. Thomson, Charles Proteus Steinmetz, Oliver Heaviside, and Charles Fortescue, when you combine Electrostatic and Magnetic Lines of Force in Faraday Tubes, you induce electricity, so Remnant Magnetism and the Reactor Rod spinning inside the intake pipe of the GEET Reactor or Dorian Demetrulias' new Hexemeter Hydrogen Reactor's design are needed to transduce the static electricity into electric current by engendering the scenario where Electrostatic Discharging is able to happen around the rod, which then leads to Faraday's Law happening on the rod, leading to Lenz's Law happening on the Reactor Pipes.